

CPSC 403: Computer Science Seminar

Week 1, September 8, 2026

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Today

- What this course is for, and what it is not
- Danny Briere: Elting Center and AI Center senior projects (about 20 minutes)
- CPSC 498: how to register and who can advise
- Two talks, three short pieces, one pitch to the faculty
- Absences, and the memo that fixes them
- Chapter 1 of *Resonate*: why most presentations fail
- Your first assignment: the "anything" talk

About me

- Trinity CS since 2022; also teaching CPSC 415, AI Integration, on Mondays
- Thirty years of technical training and conference talks: Java, Spring, Kotlin, AI
- Author of *Claude Code: Up and Running* and several other books
- Future director of the college's AI Center

Office hours: Tuesdays 10 AM to noon, MECC 175, or by appointment.

What this course is for

- The catalog says: research presentations, critical discussion, the senior exercise
- In practice: **one skill**. Presenting technical work so an audience understands it, cares, and acts
- The real deliverable is in April: a ten-minute talk on your senior project at the department's student conference
- A Travelers representative attends. Prizes go to the best talks.
- This fall builds the skills. The spring polishes the talk.

What this course is not

- Not the place your senior project happens. That is CPSC 498 with your advisor.
- Not a lecture course. Most weeks, you are the content.
- Not graded on how impressive your topic is.

A ten-minute talk about your grandmother's recipes that does the right things beats a ten-minute talk about distributed consensus that does not.

Danny Briere

Elting Center for Innovation and Entrepreneurship

CPSC 498: Senior Project Part 1

Every senior registers for CPSC 498 this fall and CPSC 499 in the spring, alongside this seminar.

1. **Talk to a faculty advisor first.** You need enough of an idea for them to say "yes, let's start." Projects change; that is expected.
2. **Then submit the special registration form** on the Registrar's site. Link on Moodle.
3. Teams of one to three. Twenty of you across two sections; the aim is twelve or fewer teams.

Who can advise this year

Faculty	Notes
Ewa Syta	Security-related projects
Nil Chakrabortii	
Jonathan Johnson	
Takunari Miyazaki	
Madalene Specialetti	
Ken Kousen	AI Center projects with government and industry partners

Peter Yoon is on leave this fall.

The shape of the semester

When	What
Weeks 1 to 2	<i>Resonate</i> chapters 1 to 3. Sign up for a talk.
Weeks 3 to 6	Presentation 1 , the "anything" talk. Chapters 4 to 7.
Week 7	Post-mortem 1. Project teams settling.
Weeks 8 to 11	Presentation 2 , the project pitch. Chapters 8, 9, coda.
Weeks 12 to 13	Post-mortem 2. Rehearsing the faculty pitch.
Finals week	Preliminary pitch to the faculty . Ten minutes per team.
No class October 13 (Trinity Days). November 24 meets.	

Two sections, one course

- Section 01 at 1:30, Section 02 at 2:55, both in this room
- Identical content, identical Moodle pages
- **Attend whichever section works in a given week.** I do not care which.
- The sign-up sheet covers both sections: three talk slots per section per week

Grading

Component	Weight
Attendance and participation	20%
Presentation 1: the "anything" talk	15%
Presentation 2: the project pitch	25%
Writing: three short pieces, pass/redo	20%
Preliminary faculty pitch	20%

Talks are graded on the *Resonate* principles, not on the topic.

Three short pieces of writing

All **pass/redo**. No numeric grade. If it is thin, it comes back once with a note.

1. **Talk analysis**, due September 22. Pick a recorded talk under twenty minutes, given to a real audience. Write the post-mortem you would write for a classmate, in Duarte's vocabulary.
2. **Post-mortem 1**, one week after your first talk. What you intended, what landed, what did not, what you will change.
3. **Post-mortem 2**, one week after your second talk. Same, plus: did you change what you said you would?

Attendance is an obligation to each other

- We meet once a week. One absence is a big slice of the course.
- After every talk, the room gives feedback. That only works if there is a room.
- If you are absent on a day a classmate presents, you took something from them.

Miss a class? Submit a one-page memo on Moodle before the next class: key points, action items, and one line that is yours: the most important takeaway, or a question you would have asked. The memo turns an unexcused absence into an excused one. No memo, it counts.

Where things live

- **Moodle**: announcements, submissions, grades, links to everything else
- **GitHub**: github.com/kousen/senior-seminar-course. Syllabus, assignment sheets, these slides.
- **Sign-up sheet**: Google Sheet linked from Moodle. First come, first served.
- **Zoom**: every class is recorded. Link on Moodle.
- **Resonate**: free online edition with registration. Link on Moodle.

Due by next week: the Academic Honesty Pledge on Moodle. Read the syllabus first.

Chapter 1

Why Resonate?

Resonance is physics before it is a metaphor

- Every object has a natural frequency. Drive it at that frequency and it moves on its own, with no contact.
- Duarte's demo: salt on a metal plate wired to a speaker. Raise the pitch and the grains reorganize into patterns, as if they knew where to go.
- Her point: **you tune the message to the audience.** The audience does not tune itself to you.
- The CS version: nobody changes their mental model because your slide said so. They change it when what you say fits something already in their head.

Why most presentations are boring

- Not because the material is dull. Because **nothing changes** for the whole time: same tone, same slide shape, same pace.
- Duarte's image: camouflage versus safety orange. Blending in is never the right move for an idea.
- Contrast is what keeps a brain tapping along: in content, in texture, in delivery.
- Her test before doing the work: *how badly do I want this idea to live?*

Facts alone fall short

- People agree with arguments and still do nothing. Being convinced is not the same as being moved.
- The objection every engineer has: "They pay me to do, not to feel."
- Duarte's answer: keep every fact. Add the reason anyone should care. Not half and half, just enough human to make the facts land.
- You have all felt this at a senior project demo. The one that worked was the one where you understood why the student cared.

Stories carry information further than bullets do

- Information is static. A story moves, so the listener moves with it.
- A story has someone who wants something, an obstacle, and a change. That is also the shape of every bug you have ever fixed.
- "We reduced latency by 40%" is a fact. "The demo froze in front of the client, and here is what we found" is a story that contains the fact.
- The uncomfortable part: good personal stories show a flaw or a failure. That is exactly why they work.

You are not the hero

- The "about me" slide at the front of most decks. The cocktail-party bore who only talks about himself.
- Duarte's rule: **the audience is the hero**. You are the mentor. Yoda, not Luke.
- A mentor has been down the road already and hands the hero what they need to get unstuck.
- What you want them thinking on the way out: "I have tools I did not have an hour ago." Not "wow, that presenter is smart."

Rule 1: Resonance causes change

Every talk in this course starts from one question:

What do I want the people in this room to believe or do when I stop talking?

If you cannot answer that in one sentence, you are not ready to build slides.

Try it now (three minutes)

Turn to the person next to you.

1. Name a topic you might use for the "anything" talk.
2. Finish this sentence out loud: "*When I stop talking, I want you to...*"
3. Swap.

Keep the sentence. It is the first line of your talk analysis, your post-mortem, and your slides.

Your first assignment: the "anything" talk

- **Ten minutes.** Alone or with one partner.
- **Any topic you care about.** A hobby, a sport, a trip, a side project, an opinion.
- Not your senior project. Teams have not formed yet, and that is the point.
- Sign up for a slot between September 22 and October 20. Three per section per week.
- Slides uploaded to Moodle before class on your day.
- Post-mortem due one week after you present.

For next week

- Read *Resonate* chapters 2 and 3
- Submit the pledge on Moodle
- Talk to a possible advisor
- Start thinking about your talk topic, and about which talk you will analyze